

API Documentation LUXORliving Version 1



Table of contents

1	General	3
2	System	4
3	Authentication	5
3.1	Login	5
3.2	Logout	5
4	Login data	6
4.1	Change Password	6
4.2	Authentication for a Request	6
5	Device	7
5.1	Diagnosis	7
6	System	8
7	Server Items	9
8	Datapoints	10
8.1	Overview	10
9	Websockets	11
10	Tunneling	12
10.1	Query current status	12
10.2	Activation/deactivation of tunneling	13
11	Link between LX project file and dp.config	14
12	LX Project File	16
12.1	View project files	16
12.2	Reading LX project data	17
12.3	Upload LX Project File	17
12.4	Delete LX Project File	17
	Contact	18

1 General

This documentation contains the API description for the LUXORliving IP1. Since the LUXORliving IP1 is almost identical to the Weinzierl BAOS 777 in terms of both software and hardware, parts of the Weinzierl BAOS 777 REST service documentation are taken over in this document and supplemented by LUXORliving-specific parts.

Any criticism, requests for improvement and changes should be addressed to Theben AG.

The following documents of Weinzierl Engineering GmbH have been linked, used or copied.

- Protocol Description: KNX BAOS REST Services
https://weinzierl.de/images/download/documents/baos/knx_ip_baos_restservices.pdf

Further documents on BAOS 777 can be found at:

https://weinzierl.de/de/developer/777-entwicklerseite/?product_post_id=388

2 System

- All components of LUXORliving communicate via a BUS connection that complies with the KNX standard (KNX S-Mode).
- The connection to the home network is made via the LUXORliving IP1 system control panel. This enables operation via the LUXORplay app.
- The system can be equipped with a maximum of 64 LUXORliving devices and includes system devices, sensors and actuators.

System Devices

- LUXORliving IP1 (system control panel),
- LUXORliving P640 (Power Supply)¹

Sensors

- Weather
- Room Sensors
- Binary inputs
- Pushbutton interfaces
- iON tactile sensors

Actuators

- Switching Actuators
- Dimming Actuators
- Venetian blind actuators
- Heating Actuators

- Operation is carried out via conventional buttons, which are integrated into the system via button interfaces/binary input. The control is always via buttons. Only for the "switch" function can you choose between "button" and "motion detector" (switch). Alternatively, operation can be carried out via the LUXORliving iON touch sensors.

Operation with the LUXORplay app for (iOS, Android, Windows PC) is also possible.

¹Only one **system control unit and one power supply may be installed** in a system.

3 Authentication

Authentication is required to access the services and API endpoints. There is a user.

3.1 Login

Request

Method	URL
POST	http://10.0.0.103/rest/login

Content Request

```
{  
    "password": "admin",  
    "username": "admin"  
}
```

Response

Status Code	200
-------------	-----

Content Response

```
3c8b531737cbd849bccf15bb9ef09d9c
```

The response must then always be sent as a cookie:

```
user=%22YOURTOKEN%22
```

3.2 Logout

Request

Method	URL
POST	http://10.0.0.103/rest/logout

Response

Status Code	204
-------------	-----

4 Login data

4.1 Change Password

Request

Method	URL
POST	http://10.0.0.103/rest/device/password

Content Request

```
{
  "credentials": {
    "current": {
      "password": "admin",
      "username": "admin"
    },
    "new": {
      "password": "newpassword",
      "username": "newuser"
    }
  }
}
```

Response

Status Code	204
-------------	-----

4.2 Authentication for a Request

Either via

1. cookie user=%22TOKEN%22
2. or the Authorization header token=TOKEN

5 Device

Request

Method	URL
POST	http://10.0.0.103/rest/device

Content Request

```
{
  "device": {
    "build_version": "379",
    "name": "NEW NAME"
  }
}
```

Response

Status Code	200
-------------	-----

5.1 Diagnosis

Request

Method	URL
GET	http://10.0.0.103/rest/device/diagnostics

Response

Status Code	200
-------------	-----

The file that is read out is encrypted and can be sent to Theben AG for further analysis.

6 System

Restart options:

- Restarting the whole device = true
- Restarting the application = false

Request

Method	URL
POST	http://10.0.0.103/rest/system/restart

Content Request

{'reboot':True/False}

Response

Status Code	200
-------------	-----

7 Server Items

Server items represent properties of the BAOS object server. A list of server items

Request

Method	URL
GET	http://10.0.0.103/rest/serveritems?count=3&start=8

Response

Status Code	200
-------------	-----

8 Datapoints

8.1 Overview

Request

Method	URL
GET	http://10.0.0.103/rest/datapoints?start=2&end=10

Response

Status Code	200
-------------	-----

9 Websockets

The device allows a websocket connection on `ws://10.0.0.100/websocket` or `wss://10.0.0.100/websocket`

Three authorization options have been implemented for this purpose:

1. via cookie `user=%22TOKEN%22`
2. via the Authorization Header `Token token=TOKEN`
3. via a query parameter, for example,
`ws://10.0.0.100/websocket?token=TOKEN`

 when implementing cross-side access via JavaScript, the query parameter must be used, as JavaScript's default websocket API cannot set headers or cookies.

10 Tunneling

Tunneling is only available to authenticated users. Tunneling must be explicitly enabled via `/rest/device/auth tunneling`. The tunneling server then accepts connections from the IP address for which the current session token has been issued.

10.1 Query current status

For a session, the current status can be queried via a GET request to `/rest/device/auth tunneling`.

Request

Method	URL
GET	<code>http://10.0.0.103/rest/device/auth tunneling</code>

Response

Status Code	200
-------------	-----

Content Response

```
{  
  "enabled": false  
}
```

10.2 Activation/deactivation of tunneling

To enable tunneling, a PUT request must be sent to `/rest/device/authntunneling`.

A timeout of the session token or logging out automatically deactivates the tunneling again. To disable tunneling directly, a PUT request can be sent with `{"enabled": false}`.

Request

Method	URL
PUT	<code>http://10.0.0.103/ rest/device/authntunneling</code>

Content Request

```
{  
  "enabled": true  
}
```

Response

Status Code	204
-------------	-----

11 Link between LX project file and dp.config

The identifier in the dpconfig file for each function is the group address:

```
"group_addresses": [  
    "5/1/248"  
]
```

The identifier in the LX project file (luxorProject.lxp) is the data point address:

```
<datapoint address="10744" role="Panik" id="_ee1c214e88f44a318f02ba7e72ae9513"/>
```

To convert the address from the LX project file to the group address for the dpconfig.json, the following steps must be performed:

1. Convert address to binary format
10744
= 0010 1001 1111 1000
2. For the first group address block, move the binary 11 to the right:
0010 1001 1111 1000 >>> 11
= 0000 0000 0000 0101
3. Convert the result to a decimal:
0000 0000 0000 0101
= 5
4. For the second group address block, the binary 8 needs to move to the right and perform a bitwise AND operation:

0010 1001 1111 1000 >>> 8
= 0000 0000 0010 1001
AND 0000 0000 0000 0000 0111
= 0000 0000 0000 0001

5. Convert the result to a decimal:
= 1
6. For the third group address block, the last 8 bits are needed. The binary address is used and a bitwise AND link is performed:
0010 1001 1111 1000
AND 0000 0000 1111 1111
= 0000 0000 1111 1000
7. Convert the result to decimals:
= 284

For example, the address for Panic is 5/1/248

12 LX Project File

12.1 View project files

Request

Method	URL
GET	http://10.0.0.103/rest/project

Response

Status Code	200
-------------	-----

Content Request

```
{
  "files": [{
    "last_mod": "2017-05-16 12:41:44",
    "name": "file2.json",
    "size": 1000000
  },
  {
    "last_mod": "2017-05-16 12:41:45",
    "name": "file3.project",
    "size": 1111111
  },
  {
    "last_mod": "2017-05-16 12:41:42",
    "name": "file1.txt",
    "size": 60000
  }
],
  "space": {
    "maximum": 20971520,
    "usage": 10.35266399383545,
    "used": 2171111
  }
}
```


12.2 Reading LX project data



The reading of the LX project file is only possible if either an LX project file has been imported beforehand or the LXliving system has been prepared for LXplay by LXplug beforehand

Request

Method	URL
GET	http://10.0.0.103/rest/project/luxorProject.lxp

Response

Status Code	200
-------------	-----

12.3 Upload LX Project File

Request

Method	URL
POST	http://10.0.0.103/rest/project/luxorProject.lxp

The upload http multipart/form-data request.

Response

Status Code	200
-------------	-----

12.4 Delete LX Project File

Request

Method	URL
DELETE	http://10.0.0.103/rest/project/luxorProject.lxp

Response

Status Code	204
-------------	-----

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